



Syahrul Fathoni Ahmad

Bachelor of Engineering
Computer Engineering
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Education

Degree	Institute	Current CGPA	Year
S.T., Teknik Komputer	Institut Teknologi Sepuluh Nopember, Surabaya	3.43/4.0	Aug 2025

Skills

- Full-Stack Development**: PHP, Laravel, Python, Go (Golang), React.js, Node.js, Express.js, Flask, API, SQL, MongoDB, Three.js, HTML/CSS
- AI & IoT Systems**: PyTorch, TensorRT, YOLO, LSTM, NLP Jetson Nano, ESP32, Arduino, C++, PyQt, RTMP/RTSP Streaming
- DevOps & Cloud**: Git, Docker, Nginx, Vercel, Azure, Apache

Work Experiences

- PT. Aldzama** Sep 2025 - Present
Software Engineer Gresik, Indonesia
 - ERP & DevOps**:
 - Set up office **VPN** and **on-prem server** environment (PC-based) for secure access and internal deployments.
 - Built and deployed an **in-house HRIS & Attendance ERP** as a **full-stack developer** using **Laravel**, designed for **~600 employees**.
 - Eliminated recurring **monthly subscription** and **per-user licensing** costs by replacing paid ERP with an internal platform; enables fast customization aligned with operational needs.
 - Rebuilt the **company profile website** from scratch using **React.js** + **Laravel backend/API**.
 - Project Manager**:
 - Leading development of the **Refractory Maintenance Dashboard** as **Project Manager** (scope, timeline, stakeholders).
- Freelance** 2022 - Present
Freelancer Hybrid
 - JAREE ITS (Academic Journal Platform)**: July 2025 - Oct 2025
 - Tools*: OJS, PHP, MySQL, CSS
 - Orchestrated a seamless, live migration of the OJS platform (v2 to v3), ensuring service continuity for **30k+ users** while upgrading a database of **597k+ records**.
 - Engineered the complete academic peer-review workflow and developed a **custom front-end theme** to enhance user experience.
 - Retrux-Shelf-Eye (AI Inventory Management System)**: Aug 2025 - Sep 2025
 - Tools*: Python, PyQt, PyTorch
 - Developed a real-time shelf stock monitoring application using Python, with a user interface built in **PyQt**.
 - Implemented an advanced object detection model (**RT-DETR**) using **SAHI** for sliced inference.
 - Engineered key software utilities based on client needs, including **multi-camera support** and dynamic directory management.
 - Padel-Clipper (IoT Video Highlight System)**: Aug 2025
 - Tools*: Python, Flask, ESP32, Express.js, PHP
 - Developed the **Flask**-based monitoring dashboard and integrated it with a team-built **Express.js** backend for a real-time video clipping service.
 - Engineered and programmed an **IoT push-button device** using **ESP32** and a custom PCB to trigger highlight captures from IP cameras.
 - Configured and deployed the entire system on a **VPS**, successfully scaling the product to serve nearly **250 active customers**.
 - 3D Container Loading Optimizer**: July 2025 - Aug 2025
 - Tools*: Python, Golang, FastAPI, React.js, Three.js

- ◊ Developed a web-based logistics tool to solve the 3D Bin Packing Problem, achieving a container fill rate of **87-89%**.
 - ◊ Researched and implemented multiple packing algorithms in Python, including **Genetic Algorithm**, BLF, and CLPTAC method to find the optimal solution.
 - ◊ Built the full-stack application with **React.js**, **FastAPI**, **Golang** and a **Three.js** canvas for 3D visualization.
- **PT. Telkom Indonesia Regional V** Feb 2024 - Apr 2024
Internship Programmer (ROC Department) Surabaya, Indonesia
 - ◊ **Automation & Data Pipeline Development:**
 - ◊ Reduced data retrieval latency by over **80%** by engineering an automated web scraping pipeline with Python and Apache Airflow.
 - ◊ Developed and deployed 2 OCR automation with validation step.
 - ◊ **System Integration & Reliability:**
 - ◊ Developed a resilient VPN auto-reconnect system (Power Automate) that effectively **eliminated connectivity downtime** and prevented work disruptions.
 - ◊ Created a suite of 7 integrated tools that reduced manual data processing tasks by **40%**.
 - **SDN Klampis Ngasem I** Aug 2022 - Jul 2023
Robotics Extracurricular Tutor Surabaya, Indonesia
 - ◊ Developed and assembled **Arduino-based Obstacle Avoidance Robots**, preparing *10+ units* for student use and aligning lessons with an existing robotics curriculum to support progressive learning.
 - ◊ Mentored students in the **BARONAS competition** by ITS Electrical Engineering, collaborating with a team of three instructors to deliver hands-on sessions, leading the team to reach the *semi-finals*.

Individual Projects

- **Smart Finance App (Personal Finance Management System)** Sep 2025
Tools: Go (Golang), React.js, Python, Scikit-learn, PostgreSQL 
 - ◊ Developed a full-stack personal finance management application that helps users track income and expenses with intelligent categorization. The system integrates a Go backend, React frontend, and a Python AI service to provide automatic transaction categorization using machine learning.
- **Padel-Clipper (IoT Video Highlight System with Golang)** June 2025 - August 2025
Tools: Go (Golang), RESTful API, Flask, Python, PostgreSQL 
 - ◊ Developed a RESTful API in Go (Golang), leveraging goroutines and channels to efficiently handle concurrent requests from multiple IoT devices for a real-time video clipping service.
- **Vehicle Speed Estimation System Using Drone with Jetson Nano-based YOLOv8 Method** Feb 2025 - May 2025
Tools: Python, YOLOv8, Jetson-Nano, TensorRT, RTMP 
 - ◊ Developed a real-time vehicle speed estimation system, achieving a peak measurement accuracy of **99.55%** and consistent performance of over 90% at an optimal 20-meter altitude.
 - ◊ Trained a superior YOLOv8 model for night-time conditions (mAP50 of **0.93**) and leveraged **TensorRT** to triple inference speed (from **>90ms** to **<35ms**), enabling real-time processing on the Jetson Nano.
- **BengkelMate: Workshop Business Process Digitalization** Oct 2024 - Jan 2025
Tools: MERN Stack (MongoDB, Express.js, React.js, Node.js) 
 - ◊ As part of a three-person team, developed a web application that digitized the workshop workflow through a role-based system for **Security, Service Advisor, and Spare Parts teams**.
 - ◊ Delivered a high-quality user experience, validated by excellent Lighthouse scores in **Performance (96)**, **Best Practices (96)**, and **SEO (92)**, ensuring a fast and reliable tool for daily operations.

Campus Organization Experiences

- **BK MWA-WM ITS** Jul 2023 - Jul 2024
Head of Human Resources Surabaya, Indonesia
 - ◊ Managed the end-to-end employee lifecycle and cultural development for **40+ staff**, encompassing recruitment (**processing 50+ applicants**), training, and employee well-being initiatives.
- **TEKMEET 2023** Jan 2023 - Mar 2023
Project Officer Surabaya, Indonesia
 - ◊ Organized sports tournament for **300+ students** and coordinated charity event serving **50+ people**.

Certifications

- **Intern – PT. Telkom Indonesia Regional V**, [Internship-Certificate](#) *Feb 2024 - Apr 2024*
- **Intermediate Machine Learning by Kaggle**, [Course](#) *Dec 2023*
- **Feature Engineering by Kaggle**, [Course](#) *Dec 2023*
- **Marketing Content with Gapura Digital Google**, [Course](#) *Sept 2021*